



# Cohort canonical evaluation — 70 agents

idea-01-lateral-attribution · 534 held-out segments · V0  
yaw 0.01627 rad/s, CTE 254.26 m · generated  
2026-06-01 17:39

- Executive summary

AGENTS (OK / TOTAL)

**69 / 70**

EVAL POOL

**534 segs**

V0 YAW RMSE

**0.01627**

V0 CTE RMSE

**254.26 m**

WALL TIME

# 36.56s

Best yaw: module-3.v3\_agent-06 **+57.3%** · Best CTE:

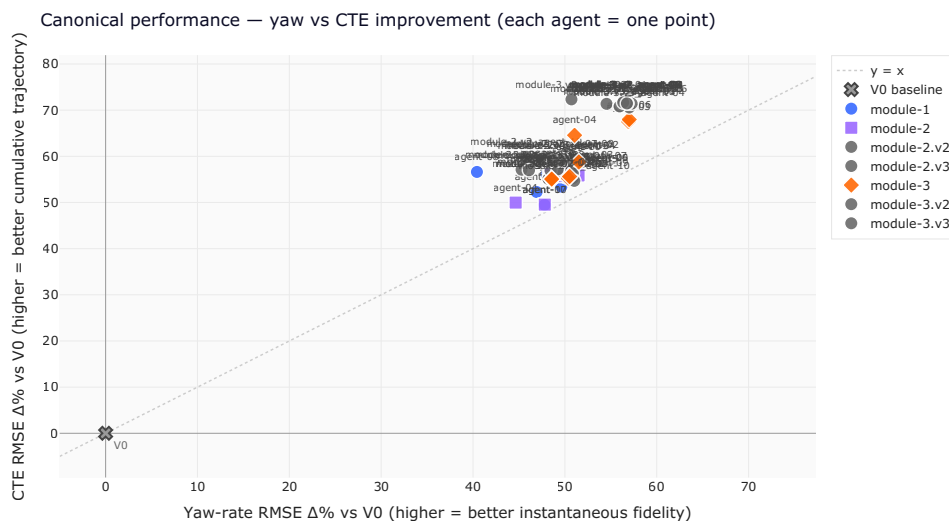
module-3.v3\_agent-03 **+72.3%**

**Winning both KPIs  $\geq$  +30% (69 agents):**

module-3.v3\_agent-06, module-3.v3\_agent-04, m3-agent-06,  
m3-agent-03, module-3.v2\_agent-04, module-3.v3\_agent-08,  
module-3.v2\_agent-09, module-3.v3\_agent-10,  
module-3.v2\_agent-02, module-3.v2\_agent-03, m3-agent-01,  
module-3.v2\_agent-05, module-3.v2\_agent-06,  
module-3.v2\_agent-07, module-3.v2\_agent-08,  
module-3.v2\_agent-10, module-3.v3\_agent-07,  
module-3.v3\_agent-09, module-3.v3\_agent-02,  
module-3.v3\_agent-01, module-3.v3\_agent-05,  
module-3.v2\_agent-01, m3-agent-10, m2-agent-05, m3-  
agent-04, module-2.v3\_agent-10, m1-agent-01,  
module-2.v2\_agent-09, module-2.v3\_agent-09,  
module-2.v3\_agent-06, module-3.v3\_agent-03,  
module-2.v3\_agent-07, m3-agent-09, m3-agent-08, m2-  
agent-08, m1-agent-09, module-2.v3\_agent-02,  
module-2.v2\_agent-05, m1-agent-07, module-2.v3\_agent-04,  
module-2.v2\_agent-08, module-2.v3\_agent-03, m2-agent-10,  
module-2.v2\_agent-04, module-2.v3\_agent-05, m1-agent-02,  
m1-agent-05, m2-agent-01, module-2.v2\_agent-06, m3-  
agent-07, m3-agent-02, module-2.v2\_agent-02, m2-agent-03,  
m1-agent-03, m2-agent-02, module-2.v3\_agent-01, m2-  
agent-09, m2-agent-06, module-2.v2\_agent-01, m1-agent-04,  
m2-agent-07, m1-agent-10, module-2.v2\_agent-07, m1-  
agent-06, module-2.v3\_agent-08, module-2.v2\_agent-10,  
module-2.v2\_agent-03, m2-agent-04, m1-agent-08

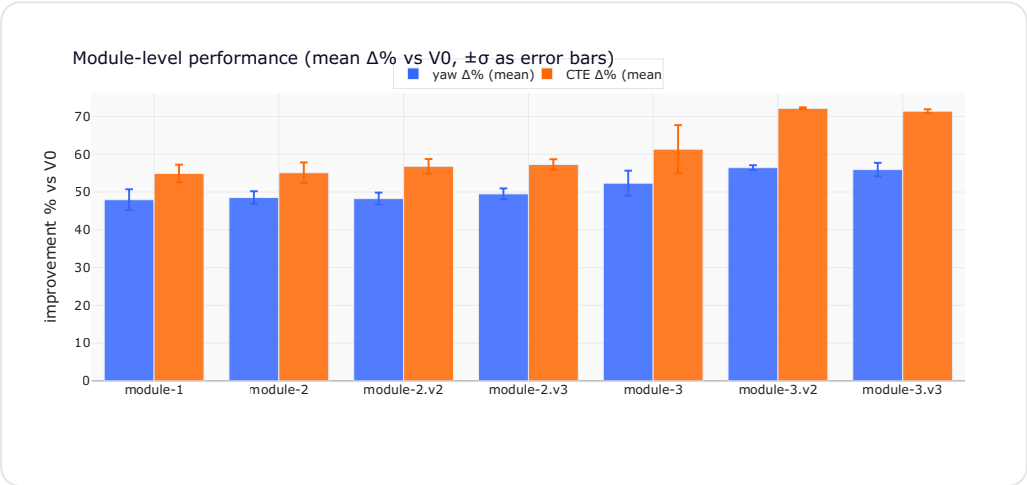
## 1. Headline scatter — yaw vs CTE improvement

Each agent is one point. **Top-right is the goal** — both KPIs improve over V0. The dashed diagonal is  $y = x$ : above it, an agent improved CTE more than yaw (good trajectory integration); below, the opposite. The V0 baseline sits at the origin.



## 2. Module-level aggregation

Performance grouped by which module shipped the agent.  
Mean  $\Delta\%$  over only the agents that reconstructed successfully;  
failures shown separately.



| family      | n ok / total | yaw $\Delta\%$ (mean $\pm \sigma$ ) | CTE $\Delta\%$ |
|-------------|--------------|-------------------------------------|----------------|
| module-1    | 10/10        | +48.0% $\pm$ 2.8%                   | +54.9          |
| module-2    | 10/10        | +48.6% $\pm$ 1.7%                   | +55.1          |
| module-2.v2 | 10/10        | +48.3% $\pm$ 1.6%                   | +56.8          |
| module-2.v3 | 10/10        | +49.5% $\pm$ 1.4%                   | +57.3          |
| module-3    | 9/10         | +52.3% $\pm$ 3.3%                   | +61.3          |
| module-3.v2 | 10/10        | +56.5% $\pm$ 0.6%                   | +72.2          |
| module-3.v3 | 10/10        | +55.9% $\pm$ 1.8%                   | +71.4          |

- 2b. Token expenditure per agent

Sourced from each agent's Claude Code subagent transcript (~/.claude/projects/<proj>/\*/subagents/agent-\*.jsonl). Tokens summed across every assistant turn of the latest-time run per agent.

COHORT TOTAL

259.43M

MEDIAN / AGENT

3.53M

MEDIAN TURNS

66

AGENTS W/ TRANSCRIPT

70

By family

| family   | n  | total tokens | median / agent | median turns |
|----------|----|--------------|----------------|--------------|
| module-1 | 10 | 23.38M       | 2.38M          | 66           |
| module-2 | 10 | 25.69M       | 2.24M          | 51           |
|          |    |              |                |              |

| family      | n  | total tokens  | median / agent | median |
|-------------|----|---------------|----------------|--------|
| module-2.v2 | 10 | <b>30.22M</b> | 3.13M          | !      |
| module-2.v3 | 10 | <b>37.55M</b> | 3.92M          | (      |
| module-3    | 10 | <b>41.38M</b> | 3.50M          | (      |
| module-3.v2 | 10 | <b>43.93M</b> | 4.49M          | (      |
| module-3.v3 | 10 | <b>57.27M</b> | 5.50M          | !      |

**Per agent (sorted by total)**

| agent                | family      | turns | total        | i |
|----------------------|-------------|-------|--------------|---|
| module-3.v3_agent-07 | module-3.v3 | 109   | <b>9.24M</b> |   |
| m3-agent-06          | module-3    | 107   | <b>8.18M</b> |   |
| module-3.v3_agent-03 | module-3.v3 | 113   | <b>7.36M</b> |   |
| module-3.v3_agent-06 | module-3.v3 | 98    | <b>7.29M</b> |   |
| m3-agent-01          | module-3    | 94    | <b>6.78M</b> |   |
| module-3.v3_agent-09 | module-3.v3 | 97    | <b>6.42M</b> |   |
| module-3.v3_agent-01 | module-3.v3 | 84    | <b>6.22M</b> |   |
| m3-agent-05          | module-3    | 106   | <b>6.03M</b> |   |
| module-3.v2_agent-03 | module-3.v2 | 81    | <b>5.52M</b> |   |
| module-3.v2_agent-06 | module-3.v2 | 78    | <b>5.16M</b> |   |
| module-3.v2_agent-02 | module-3.v2 | 76    | <b>4.89M</b> |   |
|                      |             |       |              |   |

| agent                | family      | turns | total        | i |
|----------------------|-------------|-------|--------------|---|
| module-3.v2_agent-01 | module-3.v2 | 70    | <b>4.83M</b> |   |
| module-3.v3_agent-10 | module-3.v3 | 70    | <b>4.79M</b> |   |
| module-2.v3_agent-05 | module-2.v3 | 73    | <b>4.79M</b> |   |
| module-3.v2_agent-05 | module-3.v2 | 75    | <b>4.75M</b> |   |
| module-3.v3_agent-04 | module-3.v3 | 75    | <b>4.47M</b> |   |
| m2-agent-05          | module-2    | 83    | <b>4.45M</b> |   |
| module-2.v3_agent-01 | module-2.v3 | 67    | <b>4.43M</b> |   |
| module-2.v3_agent-03 | module-2.v3 | 67    | <b>4.26M</b> |   |
| module-3.v2_agent-04 | module-3.v2 | 64    | <b>4.23M</b> |   |
| module-3.v2_agent-09 | module-3.v2 | 69    | <b>4.20M</b> |   |
| module-3.v3_agent-08 | module-3.v3 | 70    | <b>4.13M</b> |   |
| module-3.v3_agent-05 | module-3.v3 | 60    | <b>4.06M</b> |   |
| module-3.v2_agent-10 | module-3.v2 | 66    | <b>4.03M</b> |   |
| module-2.v3_agent-06 | module-2.v3 | 73    | <b>4.01M</b> |   |
| module-2.v3_agent-02 | module-2.v3 | 69    | <b>3.98M</b> |   |
| m3-agent-03          | module-3    | 72    | <b>3.96M</b> |   |
| module-2.v3_agent-04 | module-2.v3 | 67    | <b>3.86M</b> |   |
| m3-agent-08          | module-3    | 68    | <b>3.84M</b> |   |
| m2-agent-08          | module-2    | 76    | <b>3.73M</b> |   |
|                      |             |       |              |   |

| agent                | family      | turns | total        | i |
|----------------------|-------------|-------|--------------|---|
| module-3.v2_agent-07 | module-3.v2 | 58    | <b>3.68M</b> |   |
| module-2.v2_agent-07 | module-2.v2 | 56    | <b>3.63M</b> |   |
| module-2.v3_agent-07 | module-2.v3 | 69    | <b>3.61M</b> |   |
| m2-agent-07          | module-2    | 69    | <b>3.61M</b> |   |
| m1-agent-09          | module-1    | 79    | <b>3.54M</b> |   |
| module-2.v2_agent-01 | module-2.v2 | 64    | <b>3.53M</b> |   |
| module-2.v3_agent-10 | module-2.v3 | 66    | <b>3.41M</b> |   |
| module-3.v3_agent-02 | module-3.v3 | 70    | <b>3.29M</b> |   |
| module-2.v2_agent-04 | module-2.v2 | 61    | <b>3.29M</b> |   |
| module-2.v2_agent-08 | module-2.v2 | 59    | <b>3.19M</b> |   |
| m3-agent-07          | module-3    | 66    | <b>3.16M</b> |   |
| module-2.v2_agent-03 | module-2.v2 | 58    | <b>3.14M</b> |   |
| module-2.v2_agent-02 | module-2.v2 | 54    | <b>3.11M</b> |   |
| m3-agent-09          | module-3    | 61    | <b>3.02M</b> |   |
| module-2.v2_agent-09 | module-2.v2 | 57    | <b>3.00M</b> |   |
| m1-agent-08          | module-1    | 74    | <b>2.96M</b> |   |
| module-2.v2_agent-05 | module-2.v2 | 54    | <b>2.88M</b> |   |
| m3-agent-10          | module-3    | 65    | <b>2.86M</b> |   |
| m3-agent-02          | module-3    | 61    | <b>2.83M</b> |   |
|                      |             |       |              |   |

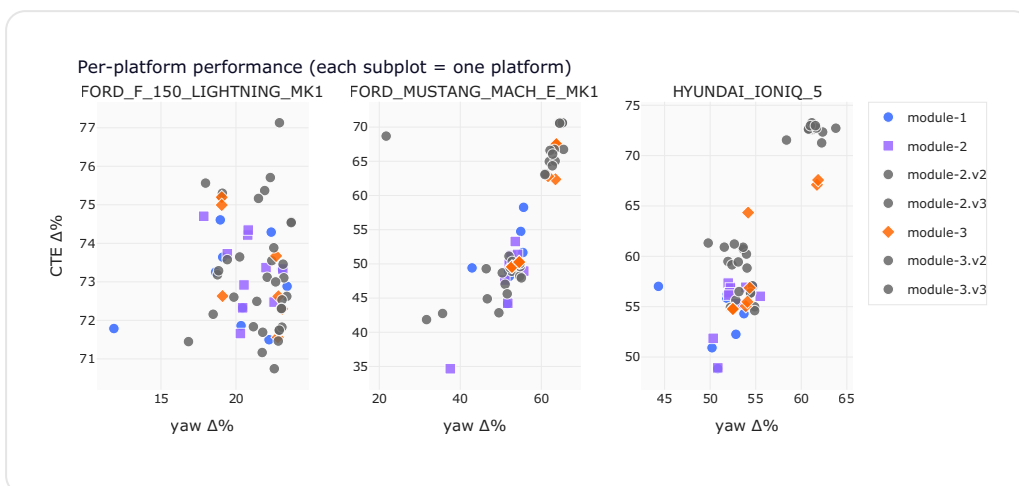


| agent                | family      | turns | total        | i |
|----------------------|-------------|-------|--------------|---|
| module-2.v3_agent-09 | module-2.v3 | 56    | <b>2.72M</b> |   |
| m2-agent-02          | module-2    | 68    | <b>2.68M</b> |   |
| m1-agent-02          | module-1    | 66    | <b>2.66M</b> |   |
| module-3.v2_agent-08 | module-3.v2 | 47    | <b>2.63M</b> |   |
| module-2.v3_agent-08 | module-2.v3 | 55    | <b>2.49M</b> |   |
| m1-agent-10          | module-1    | 64    | <b>2.45M</b> |   |
| m1-agent-07          | module-1    | 62    | <b>2.41M</b> |   |
| module-2.v2_agent-06 | module-2.v2 | 48    | <b>2.37M</b> |   |
| m2-agent-03          | module-2    | 52    | <b>2.35M</b> |   |
| m1-agent-04          | module-1    | 63    | <b>2.34M</b> |   |
| m1-agent-05          | module-1    | 62    | <b>2.25M</b> |   |
| m2-agent-09          | module-2    | 55    | <b>2.14M</b> |   |
| m2-agent-10          | module-2    | 53    | <b>2.11M</b> |   |
| module-2.v2_agent-10 | module-2.v2 | 42    | <b>2.07M</b> |   |
| m1-agent-01          | module-1    | 53    | <b>1.88M</b> |   |
| m2-agent-04          | module-2    | 45    | <b>1.83M</b> |   |
| m1-agent-06          | module-1    | 54    | <b>1.73M</b> |   |
| m2-agent-06          | module-2    | 43    | <b>1.66M</b> |   |
| m1-agent-03          | module-1    | 38    | <b>1.17M</b> |   |
|                      |             |       |              |   |

| agent       | family   | turns | total         | i |
|-------------|----------|-------|---------------|---|
| m2-agent-01 | module-2 | 41    | <b>1.13M</b>  |   |
| m3-agent-04 | module-3 | 26    | <b>740.2k</b> |   |

### • 3. Per-platform breakdown

Same cohort, but split by the vehicle platform the agent was evaluated on. Agents that win one platform but lose another have a platform-specific calibration; agents whose dots cluster together generalise.

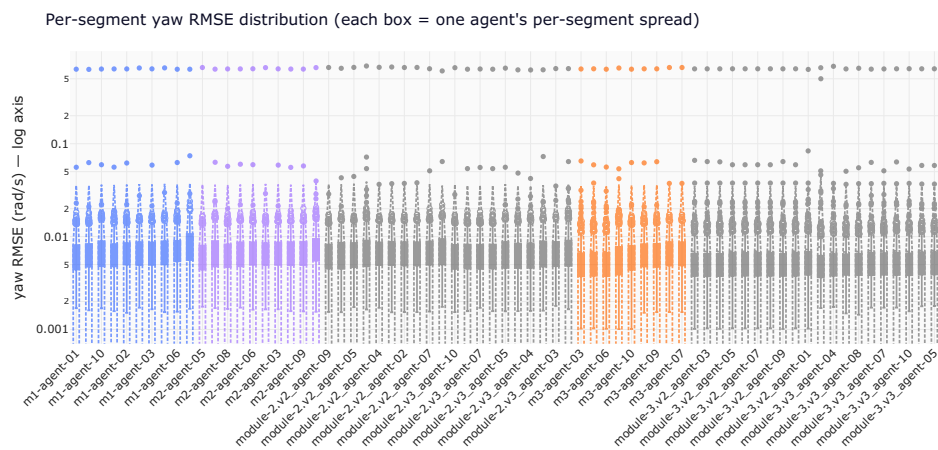


| platform                 | agents | yaw $\Delta\%$ (mean $\pm \sigma$ ) |
|--------------------------|--------|-------------------------------------|
| FORD_F_150_LIGHTNING_MK1 | 69     | +21.3% $\pm$ 2.1%                   |
| FORD_MUSTANG_MACH_E_MK1  | 69     | +55.2% $\pm$ 8.2%                   |

| platform        | agents | yaw $\Delta\%$ (mean $\pm \sigma$ ) |
|-----------------|--------|-------------------------------------|
| HYUNDAI_IONIQ_5 | 69     | +55.6% $\pm$ 4.3%                   |

## • 4. Per-segment yaw RMSE distribution

Each box summarises one agent's per-segment yaw RMSE — pooled RMSE can hide that an agent does great on most segments but pathologically badly on a few. The y-axis is logarithmic to compress outliers.



## ● 5. Per-agent canonical scorecard

The full table. Failed reconstructions are surfaced with their reason — no fake zeros.

| agent       | family   | status |
|-------------|----------|--------|
| m1-agent-01 | module-1 | ok     |
| m1-agent-02 | module-1 | ok     |
| m1-agent-03 | module-1 | ok     |
| m1-agent-04 | module-1 | ok     |
| m1-agent-05 | module-1 | ok     |
| m1-agent-06 | module-1 | ok     |
| m1-agent-07 | module-1 | ok     |
| m1-agent-08 | module-1 | ok     |
| m1-agent-09 | module-1 | ok     |
| m1-agent-10 | module-1 | ok     |
| m2-agent-01 | module-2 | ok     |
| m2-agent-02 | module-2 | ok     |
| m2-agent-03 | module-2 | ok     |

| agent                | family      | status |
|----------------------|-------------|--------|
|                      |             |        |
| m2-agent-04          | module-2    | ok     |
| m2-agent-05          | module-2    | ok     |
| m2-agent-06          | module-2    | ok     |
| m2-agent-07          | module-2    | ok     |
| m2-agent-08          | module-2    | ok     |
| m2-agent-09          | module-2    | ok     |
| m2-agent-10          | module-2    | ok     |
| module-2.v2_agent-01 | module-2.v2 | ok     |
| module-2.v2_agent-02 | module-2.v2 | ok     |
| module-2.v2_agent-03 | module-2.v2 | ok     |
| module-2.v2_agent-04 | module-2.v2 | ok     |
| module-2.v2_agent-05 | module-2.v2 | ok     |
| module-2.v2_agent-06 | module-2.v2 | ok     |
| module-2.v2_agent-07 | module-2.v2 | ok     |
| module-2.v2_agent-08 | module-2.v2 | ok     |
| module-2.v2_agent-09 | module-2.v2 | ok     |
|                      |             |        |

| agent                | family      | status        |
|----------------------|-------------|---------------|
| module-2.v2_agent-10 | module-2.v2 | ok            |
| module-2.v3_agent-01 | module-2.v3 | ok            |
| module-2.v3_agent-02 | module-2.v3 | ok            |
| module-2.v3_agent-03 | module-2.v3 | ok            |
| module-2.v3_agent-04 | module-2.v3 | ok            |
| module-2.v3_agent-05 | module-2.v3 | ok            |
| module-2.v3_agent-06 | module-2.v3 | ok            |
| module-2.v3_agent-07 | module-2.v3 | ok            |
| module-2.v3_agent-08 | module-2.v3 | ok            |
| module-2.v3_agent-09 | module-2.v3 | ok            |
| module-2.v3_agent-10 | module-2.v3 | ok            |
| m3-agent-01          | module-3    | ok            |
| m3-agent-02          | module-3    | ok            |
| m3-agent-03          | module-3    | ok            |
| m3-agent-04          | module-3    | ok            |
| m3-agent-05          | module-3    | import_failed |
| m3-agent-06          | module-3    | ok            |
|                      |             |               |

| agent                | family      | status |
|----------------------|-------------|--------|
| m3-agent-07          | module-3    | ok     |
| m3-agent-08          | module-3    | ok     |
| m3-agent-09          | module-3    | ok     |
| m3-agent-10          | module-3    | ok     |
| module-3.v2_agent-01 | module-3.v2 | ok     |
| module-3.v2_agent-02 | module-3.v2 | ok     |
| module-3.v2_agent-03 | module-3.v2 | ok     |
| module-3.v2_agent-04 | module-3.v2 | ok     |
| module-3.v2_agent-05 | module-3.v2 | ok     |
| module-3.v2_agent-06 | module-3.v2 | ok     |
| module-3.v2_agent-07 | module-3.v2 | ok     |
| module-3.v2_agent-08 | module-3.v2 | ok     |
| module-3.v2_agent-09 | module-3.v2 | ok     |
| module-3.v2_agent-10 | module-3.v2 | ok     |
| module-3.v3_agent-01 | module-3.v3 | ok     |
| module-3.v3_agent-02 | module-3.v3 | ok     |
| module-3.v3_agent-03 | module-3.v3 | ok     |
|                      |             |        |

| agent                | family      | status |
|----------------------|-------------|--------|
| module-3.v3_agent-04 | module-3.v3 | ok     |
| module-3.v3_agent-05 | module-3.v3 | ok     |
| module-3.v3_agent-06 | module-3.v3 | ok     |
| module-3.v3_agent-07 | module-3.v3 | ok     |
| module-3.v3_agent-08 | module-3.v3 | ok     |
| module-3.v3_agent-09 | module-3.v3 | ok     |
| module-3.v3_agent-10 | module-3.v3 | ok     |

- **6. Calibration cards — fitted coefficients**

Where the cohort converges on similar physics, and where it forks. Each tile is one agent's `coeffs.json` as shipped.

m1-agent-02

```
{
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.4760851016953556,
    "K_us": 0.0026722138032651867,
```



```
    "bias_rad": 0.0002686192499419401
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.809760457416018,
    "K_us": 0.0038412171248315657,
    "bias_rad": 0.001307379955568698
  },
  "HYUNDAI_IONIQ_5": {
    "L": 3.096229822470566,
    "K_us": 0.003695564361835472,
    "bias_rad": -0.0002960976826300522
  },
  "TESLA_MODEL_3": {
    "L": 2.875,
    "K_us": 0.0,
    "bias_rad": 0.0
  }
}
```

---

## m1-agent-04

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "a": -0.004329037631466362,
    "b": 0.9177897626357783,
    "c": -0.00044178131403551616
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "a": 0.0002206757945255914,
    "b": 1.1513559927303039,
    "c": -0.0005583363495612761
  },
  "HYUNDAI_IONIQ_5": {
    "a": 0.0020222989265629857,
```

```
    "b": 0.9120045284217976,  
    "c": -0.00048449411444108955  
  },  
  "TESLA_MODEL_3": {  
    "a": 0.0002206757945255914,  
    "b": 1.1513559927303039,  
    "c": -0.0005583363495612761  
  },  
  "_default": {  
    "a": 0.0,  
    "b": 1.0,  
    "c": 0.0  
  }  
}
```

---

## m1-agent-06

```
{  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "L": 2.984,  
    "best": "M2_understeer",  
    "m1": {  
      "a": 1.0870612234849137  
    },  
    "m2": {  
      "L_eff": 2.4763030468069056,  

```

```

    "K_us": 0.0024395167795073017
  },
  "m3": {
    "a": 0.9751496691433119,
    "b": 0.23839497273290036,
    "c": 0.0004082870644824788
  },
  "val_rmse": {
    "V0": 0.0267999381028469,
    "M1": 0.027439581036525294,
    "M2": 0.026420879339395156,
    "M3": 0.031903168689021696
  },
  "train_rmse": {
    "V0": 0.012685628695945105,
    "M1": 0.011389013310887795,
    "M2": 0.008745921115146504,
    "M3": 0.010386863485176999
  }
},
"FORD_F_150_LIGHTNING_MK1": {
  "L": 3.7,
  "best": "M2_understeer",
  "m1": {
    "a": 0.8569324797501493
  },
  "m2": {
    "L_eff": 3.882198880475702,
    "K_us": 0.003439279945738158
  },
  "m3": {
    "a": 0.8772240685582882,
    "b": -0.031353830652592465,
    "c": -0.0036354381262862774
  },
  "val_rmse": {
    "V0": 0.01591104278075153,
    "M1": 0.011141896721881754,
    "M2": 0.0073696198273352906,
    "M3": 0.0104650790814036
  },
  "train_rmse": {

```

```
"V0": 0.02018452928972004,
```

## m1-agent-08

```
{
  "HYUNDAI_IONIQ_5": {
    "L": 3.0,
    "K": 0.0033895612010179177,
    "s": 0.9668278221152291,
    "off": 0.0005045046778856075,
    "lag": 3,
    "rmse": 0.00888482422851739
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "L": 2.984,
    "K": 0.002625595512351926,
    "s": 1.1771921499482927,
    "off": 3.9923690213217876e-05,
    "lag": 4,
    "rmse": 0.01315660341959157
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "L": 3.7,
    "K": 0.0035388455508032707,
    "s": 0.9569361755429138,
    "off": -0.0011908292677060655,
    "lag": 3,
    "rmse": 0.011832340064259452
  },
  "TESLA_MODEL_3": {
    "L": 2.875,
    "K": 0.0033895612010179177,
    "s": 1.0,
    "off": 0.0,
    "lag": 3,
    "rmse": null
  }
}
```

```
}  
}
```

## m1-agent-10

```
{  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "L_eff": 2.5297190644305063,  
    "K_us": 0.0015446536838391765,  
    "d0": -4.756112262525493e-05,  
    "tau_yaw": 0.02,  
    "tau_steer": 0.05  
  },  
  "FORD_F_150_LIGHTNING_MK1": {  
    "L_eff": 3.834827559665228,  
    "K_us": 0.0031524889056087045,  
    "d0": 0.00115199881391885,  
    "tau_yaw": 0.0,  
    "tau_steer": 0.05  
  },  
  "HYUNDAI_IONIQ_5": {  
    "L_eff": 3.1485752690608293,  
    "K_us": 0.0021732037046845788,  
    "d0": -0.0005619614330402095,  
    "tau_yaw": 0.0,  
    "tau_steer": 0.05  
  },  
  "TESLA_MODEL_3": {  
    "_note": "no truth available \u2014 passthrough yaw_rate_pred.  
    "passthrough": true  
  }  
}
```

## m2-agent-03

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "platform": "FORD_F_150_LIGHTNING_MK1",
    "L": 3.7,
    "n_samples": 430990,
    "rmse_v0": 0.016327157090019005,
    "rmse_v1": 0.007867193883977044,
    "K_v1": 0.0043769531249999985,
    "rmse_v2": 0.00640773025816659,
    "K_v2": 0.003836687017912508,
    "a_v2": 0.9743406550510199,
    "b_v2": -0.001228592917784131
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "platform": "FORD_MUSTANG_MACH_E_MK1",
    "L": 2.984,
    "n_samples": 615175,
    "rmse_v0": 0.013616820108542612,
    "rmse_v1": 0.013930807470187448,
    "K_v1": 0.000785400390624989,
```

```

    "rmse_v2": 0.0095149036095479,
    "K_v2": 0.0028742800060782272,
    "a_v2": 1.1998919734409932,
    "b_v2": 3.385609342398125e-05
  },
  "HYUNDAI_IONIQ_5": {
    "platform": "HYUNDAI_IONIQ_5",
    "L": 3.0,
    "n_samples": 2042270,
    "rmse_v0": 0.017084454875116246,
    "rmse_v1": 0.009100616867655459,
    "K_v1": 0.0042265624999999985,
    "rmse_v2": 0.008741269113389604,
    "K_v2": 0.0033859035164445724,
    "a_v2": 0.9707125684779052,
    "b_v2": 0.0005066770839987745
  }
}

```

## m2-agent-07

```

{
  "FORD_F_150_LIGHTNING_MK1": {
    "tau": 0.05,
    "L_eff": 3.8588781722695416,
    "K_u": 0.00314039574481963,
    "delta_bias_rad": -0.001157331673169034,
    "rmse": 0.005592721659455879
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "tau": 0.08,
    "L_eff": 2.5486304264900417,
    "K_u": 0.0015312706121448161,
    "delta_bias_rad": 4.8217638587095315e-05,
    "rmse": 0.008347407485517062
  },
}

```

```
"HYUNDAI_IONIQ_5": {  
  "tau": 0.05,  
  "L_eff": 3.163750474852756,  
  "K_u": 0.0021607848879418314,  
  "delta_bias_rad": 0.0005617097326308059,  
  "rmse": 0.007713638558960502  
}  
}
```

---

### m3-agent-01

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "g": 0.863,  
    "L_eff": 3.26,  
    "K_us": 0.0035,  
    "tau": 0.06,
```



```

    "delta0": 0.00133,
    "use_per_segment_delta0": false
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 0.891,
    "L_eff": 2.22,
    "K_us": 0.0015,
    "tau": 0.069,
    "delta0": -0.0001,
    "use_per_segment_delta0": true
  },
  "HYUNDAI_IONIQ_5": {
    "g": 0.93817,
    "L_eff": 2.8871,
    "K_us": 0.0028925,
    "tau": 0.061895,
    "delta0": 0.0,
    "use_per_segment_delta0": true
  },
  "_meta": {
    "delta0_detector": {
      "method": "yaw_rate_pred_rads_low_speed_high",
      "yr_thresh": 0.03,
      "v_thresh": 5.0,
      "min_rows": 50
    },
    "notes": "Lightning coeffs hand-set from anti-patterns.md prior"
  }
}

```

### m3-agent-03

```

{
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 1.031539710821034,
    "L_eff": 2.605202152510245,
    "K_us": 0.001860730244989818,
    "tau": 0.0627605678289049,

```

```

    "delta0_fallback": 0.00016924506921824425,
    "v_thresh": 8.0,
    "delta_thresh": 0.005,
    "min_rows": 80,
    "use_per_segment_delta0": true
  },
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.8860740528359514,
    "L_eff": 3.380345065967063,
    "K_us": 0.0032867081270640908,
    "tau": 0.05844688797895399,
    "delta0": 0.001336241196109372,
    "use_per_segment_delta0": false
  },
  "HYUNDAI_IONIQ_5": {
    "g": 0.896091628794913,
    "L_eff": 2.8008769715764292,
    "K_us": 0.0023750439062653513,
    "tau": 0.04533958799360384,
    "delta0_fallback": -0.00028615565156215293,
    "v_thresh": 8.0,
    "delta_thresh": 0.005,
    "min_rows": 80,
    "use_per_segment_delta0": true
  }
}

```

---

### m3-agent-06

```

{
  "FORD_F_150_LIGHTNING_MK1": {
    "use_per_segment_delta0": false,
    "delta0": 0.0010739940929899243,
    "g": 0.8428393582368884,
    "L_eff": 3.26,
    "K_us": 0.0024711101464802783,
    "tau": 0.05493261284467375
  },

```

```
"FORD_MUSTANG_MACH_E_MK1": {
  "use_per_segment_delta0": true,
  "delta0_fallback": -9.587089484890604e-05,
  "delta0": -9.587089484890604e-05,
  "g": 0.8814740811859291,
  "L_eff": 2.22,
  "K_us": 0.001600317906539393,
  "tau": 0.08022498681872611
},
"HYUNDAI_IONIQ_5": {
  "use_per_segment_delta0": true,
  "delta0_fallback": 7.704255037895423e-05,
  "delta0": 7.704255037895423e-05,
  "g": 0.9863062253350734,
  "L_eff": 3.033023238719065,
  "K_us": 0.0028341308076772504,
  "tau": 0.055317906584940124
},
"TESLA_MODEL_3": {
  "passthrough": true
}
}
```

---

### m3-agent-08

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.6006343437133439,
    "delta0": 0.0012709657890603524,
    "K_us": 0.002411110430682984,
    "tau": 0.05905789911423279,
```

```

    "L_eff": 2.273362244343053,
    "loss": 0.005658956187762647,
    "n_segments_fit": 175
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 0.9449514060826516,
    "delta0": -2.802364842758471e-05,
    "K_us": 0.0023030236477064245,
    "tau": 0.06905315476018789,
    "L_eff": 2.3418351435366787,
    "loss": 0.008957550273019014,
    "n_segments_fit": 240
  },
  "HYUNDAI_IONIQ_5": {
    "g": 0.8635558962089673,
    "delta0": -0.0002751563706215791,
    "K_us": 0.0031040456441715804,
    "tau": 0.051877028201724164,
    "L_eff": 2.663564560241335,
    "loss": 0.009441409401572077,
    "n_segments_fit": 400
  }
}

```

## m3-agent-10

```

{
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 1.2677013760203653,
    "L_eff": 4.8869672279395875,
    "K_us": 0.004638743872488176,
    "tau": 0.056604854781974856,
    "delta0_glob": 0.0012307692372779105,
    "use_per_segment_delta0": false,
    "train_yaw_rmse": 0.012681866822813753
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 1.4999996985088524,

```

```
    "L_eff": 3.7976856683656175,  
    "K_us": 0.0030911056729474493,  
    "tau": 0.06768316418910694,  
    "delta0_glob": 0.0003094591239113909,  
    "use_per_segment_delta0": true,  
    "train_yaw_rmse": 0.013524463171650382  
  },  
  "HYUNDAI_IONIQ_5": {  
    "g": 1.4070879006714407,  
    "L_eff": 4.3500385986664405,  
    "K_us": 0.004915645105647427,  
    "tau": 0.05040893663177004,  
    "delta0_glob": -0.0005219437645709744,  
    "use_per_segment_delta0": false,  
    "train_yaw_rmse": 0.009412230675873393  
  }  
}
```

---

## module-2.v2\_agent-02

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "k": 0.9366323579288449,  
    "K_us": 0.0008734759361408742,  
    "b": -0.004441819259140514  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "k": 1.1760967499879564,  
    "K_us": 0.0008734571660548825,  
    "b": 0.0001733616708132899  
  },  
  "HYUNDAI_IONIQ_5": {  
    "k": 0.9342137556967759,  
    "K_us": 0.0009874770996559546,  
    "b": 0.002002086108887664  
  },  
  "TESLA_MODEL_3": {  
    "k": 1.0,
```

```
"K_us": 0.0,  
"b": 0.0  
}  
}
```

---

## module-2.v2\_agent-04

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "L": 3.7,  
    "d0": 0.0012435085448964291,  
    "kk": 0.9364948189665847,  
    "KK": 0.0032470115980856545,  
    "tau": 0.0  
  },  
  "HYUNDAI_IONIQ_5": {  
    "L": 2.97,  
    "d0": -0.0005236657821362199,  
    "kk": 0.933574693892375,  
    "KK": 0.0029030763922389792,  
    "tau": 0.0  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "L": 2.984,  
    "d0": -2.675943355525215e-05,  
    "kk": 1.1761809849040472,  
    "KK": 0.0026070689183535123,  
    "tau": 0.0  
  },  
  "TESLA_MODEL_3": {  
    "L": 2.875,  
    "d0": 0.0,  
    "kk": 1.0,  
    "KK": 0.0,  
    "tau": 0.0  
  }  
}
```

## module-2.v2\_agent-06

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "a": 0.9397533271985806,
    "K": 0.0008518149458872777,
    "b": -0.0044345248756529525
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "a": 1.1808972441892185,
    "K": 0.0008794604321610935,
    "b": 0.00016328778675285131
  },
  "HYUNDAI_IONIQ_5": {
    "a": 0.9350770580282316,
    "K": 0.0009883221917463027,
    "b": 0.002004778269326024
  },
  "TESLA_MODEL_3": {
    "a": 1.0,
    "K": 0.0,
    "b": 0.0
  }
}
```

## module-2.v2\_agent-08

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "G": 0.9041237868107276,
    "Kus": 0.0007955703708621087,
    "bias": -0.005088051550305564,
    "tau": 0.05804217686893913,
    "kdot": -0.012302023150477267
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "G": 1.1560312029550013,
    "Kus": 0.0010408901778361697,
    "bias": 0.0007331723308643077,
    "tau": 0.040688651274880756,
    "kdot": 0.008800778388725197
  },
  "HYUNDAI_IONIQ_5": {
    "G": 0.9700159639975359,
    "Kus": 0.001582087902935487,
    "bias": 0.00211745029940103,
    "tau": 0.04998898336470954,
    "kdot": 3.516097849258846e-05
  },
  "TESLA_MODEL_3": {
    "G": 1.0,
    "Kus": 0.0,
    "bias": 0.0,
    "tau": 0.0,
    "kdot": 0.0
  }
}
```



## module-2.v2\_agent-10

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "k": 0.0009299897686168917,
    "g": -0.13146377934999154,
    "b": -0.004858365647164137
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "k": 0.0007776286604362939,
    "g": 0.1760123050495998,
    "b": 0.0007151284926824161
  },
  "HYUNDAI_IONIQ_5": {
    "k": 0.0017086888155224455,
    "g": -0.02937372993329363,
    "b": 0.0030989091566628654
  },
  "TESLA_MODEL_3": {
    "k": 0.0,
    "g": 0.0,
    "b": 0.0
  }
}
```

---

## module-2.v3\_agent-02

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "L_eff": 3.941941906538264,
    "K_us": 0.0037788233688761494,
    "b": -0.005450661568006636,
    "tau": -0.08207658150714123
  },
  "FORD_MUSTANG_MACH_E_MK1": {
```

```
"L_eff": 2.5622977171066155,  
"K_us": 0.002745633390344498,  
"b": 0.00039676132180812505,  
"tau": -0.02283428861943423  
},  
"HYUNDAI_IONIQ_5": {  
  "L_eff": 3.0387877407606987,  
  "K_us": 0.004732031059157526,  
  "b": 0.0019402727789656954,  
  "tau": -0.038122141286844304  
},  
"TESLA_MODEL_3": null  
}
```

---

## module-2.v3\_agent-04

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "L_eff": 3.96394551474604,  
    "Kus": 0.003414866921617812,  
    "bias": -0.005292773687032753,  
    "tau": -0.06157679065853846  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "L_eff": 2.569483805164948,  
    "Kus": 0.003414866921617812,  
    "bias": -0.005292773687032753,  
    "tau": -0.06157679065853846  
  },  
  "TESLA_MODEL_3": null  
}
```

```
    "Kus": 0.0025788290081397765,  
    "bias": 0.0011230859100703557,  
    "tau": -0.04709908110775255  
  },  
  "HYUNDAI_IONIQ_5": {  
    "L_eff": 3.008377713685593,  
    "Kus": 0.005133658670161866,  
    "bias": 0.0019487320168033527,  
    "tau": -0.04867295794401198  
  },  
  "TESLA_MODEL_3": {  
    "dummy": 0.0  
  }  
}
```

---

## module-2.v3\_agent-06

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "platform": "FORD_F_150_LIGHTNING_MK1",  
    "L": 3.818759956786134,  
    "Kus": 0.0039432563422251285,  
    "tau": -0.05861405092290705,  
    "bias": -0.004437268567086911,  
    "rmse_fit": 0.0054979371675811595,  
    "n": 430990,  
    "converged": true  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "platform": "FORD_MUSTANG_MACH_E_MK1",  
    "L": 2.4997194805464074,  
    "Kus": 0.0024024802542751437,  
    "tau": -0.06630813663570458,  
    "bias": 0.00022227158439854945,  
    "rmse_fit": 0.00894166094428278,  
  }  
}
```

```
    "n": 615175,  
    "converged": true  
  },  
  "HYUNDAI_IONIQ_5": {  
    "platform": "HYUNDAI_IONIQ_5",  
    "L": 3.100074393615448,  
    "Kus": 0.0035463766130646316,  
    "tau": -0.05358525493540066,  
    "bias": 0.001966636983251082,  
    "rmse_fit": 0.008266374104865987,  
    "n": 2042270,  
    "converged": true  
  }  
}
```

---

## module-2.v3\_agent-08

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "L": 3.7,  
    "K_us": 0.004622784347956018,  
    "tau": -0.05460138412592134,  
    "bias": -0.004482561615678892,  
    "train_mse": 3.3520085701320475e-05,  
    "train_rmse": 0.005789653331704799,  
    "n_samples": 430990,  
    "n_segments": 164,  
    "converged": true,  
    "message": "CONVERGENCE: NORM OF PROJECTED GRADIENT <= PGTOL"  
  },  
}
```

```
"FORD_MUSTANG_MACH_E_MK1": {
  "L": 2.984,
  "K_us": 0.0008351228965168481,
  "tau": -0.09459794094766563,
  "bias": 0.00054055263346224,
  "train_mse": 0.00017605613840148047,
  "train_rmse": 0.01326861478834473,
  "n_samples": 615175,
  "n_segments": 232,
  "converged": true,
  "message": "CONVERGENCE: NORM OF PROJECTED GRADIENT <= PGTOL"
},
"HYUNDAI_IONIQ_5": {
  "L": 2.95,
  "K_us": 0.004646731366208564,
  "tau": -0.04946605447751251,
  "bias": 0.0018472769568295111,
  "train_mse": 7.578893405982132e-05,
  "train_rmse": 0.008705684008727937,
  "n_samples": 2042270,
  "n_segments": 785,
  "converged": true,
  "message": "CONVERGENCE: NORM OF PROJECTED GRADIENT <= PGTOL"
}
}
```

---

## module-2.v3\_agent-10

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "s_d": 0.9300245172021879,
    "c_d": 0.347294150023435,
    "tau_d": -0.05773764830098028,
    "K_us": 0.003240471803333766,
    "b": -0.004299834214218103,
    "L": 3.7
  },
  "FORD_MUSTANG_MACH_E_MK1": {
```

```
"s_d": 1.1508056993666387,  
"c_d": 0.7498585082419895,  
"tau_d": -0.07607250144234215,  
"K_us": 0.0022785419555576775,  
"b": 0.0003167396785048121,  
"L": 2.984  
},  
"HYUNDAI_IONIQ_5": {  
  "s_d": 0.9422644146648296,  
  "c_d": 0.3465308670029774,  
  "tau_d": -0.050503079814885346,  
  "K_us": 0.002862544790766583,  
  "b": 0.0022556662651586025,  
  "L": 3.0  
},  
"TESLA_MODEL_3": {  
  "passthrough": true  
}  
}
```

---

## module-3.v2\_agent-02

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "g": 0.8599716540800916,  
    "L_eff": 3.260802128118046,  
    "K_us": 0.0033605485538025595,  
    "tau": 0.058798336746219135,  
    "delta0": 0.0012445223286377857,  
    "use_per_segment_delta0": false  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "g": 0.8948751688442338,  
    "L_eff": 2.2184748889099999,  
    "K_us": 0.002002066992546276,  
    "tau": 0.06321825538415474,  
    "delta0": -0.0001,  
  }  
}
```

```
    "use_per_segment_delta0": true
  },
  "HYUNDAI_IONIQ_5": {
    "g": 0.928157495232038,
    "L_eff": 2.890194195032293,
    "K_us": 0.002716802199963435,
    "tau": 0.051288867659338114,
    "delta0": 0.0005262026839657872,
    "use_per_segment_delta0": true
  }
}
```

---

## module-3.v2\_agent-04

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "g": 0.8627559173927216,
    "L_eff": 3.2661642837279934,
    "K_us": 0.003399629365556295,
    "tau": 0.05764217451751205,
    "delta0": 0.0011797692301086372,
    "delta0_fallback": 0.0011797692301086372,
    "use_per_segment_delta0": false,
    "train_rmse": 0.005285188367801501,
    "dev_rmse": 0.006580546885659461,
    "bias_spread_std": null
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "g": 1.2845766366025777,
    "L_eff": 3.185078856835661,
    "K_us": 0.002783912493950955,
    "tau": 0.06245560987504152,
    "delta0": -0.0019790637549735985,
    "delta0_fallback": -0.0019790637549735985,
    "use_per_segment_delta0": true,
    "train_rmse": 0.008444302053948288,
    "dev_rmse": 0.008286227301504138,
    "bias_spread_std": null
  }
}
```

```
},
"HYUNDAI_IONIQ_5": {
  "g": 0.9454031576502736,
  "L_eff": 2.9345636591691626,
  "K_us": 0.0028198034203513055,
  "tau": 0.0507163687871907,
  "delta0": 0.0004041780050478772,
  "delta0_fallback": 0.0004041780050478772,
  "use_per_segment_delta0": true,
  "train_rmse": 0.007930751971921808,
  "dev_rmse": 0.00660747263955304,
  "bias_spread_std": null
}
}
```

---

## module-3.v2\_agent-06

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "use_per_segment_delta0": false,
    "delta0": 0.00133,
    "g": 0.863,
    "L_eff": 3.26,
    "K_us": 0.0035,
    "tau": 0.06
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "use_per_segment_delta0": true,
    "delta0_fallback": -0.0001,
    "g": 0.891,
    "L_eff": 2.22,
    "K_us": 0.0015,
    "tau": 0.069
  },
  "HYUNDAI_IONIQ_5": {
    "use_per_segment_delta0": true,
    "delta0_fallback": 0.0,
    "g": 0.938,
```



```
"L_eff": 2.887,  
"K_us": 0.00289,  
"tau": 0.062  
}  
}
```

---

### module-3.v2\_agent-08

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "use_per_segment_delta0": false,  
    "delta0": 0.00133,  
    "g": 0.863,  
    "L_eff": 3.26,  
    "K_us": 0.0035,  
    "tau": 0.06  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "use_per_segment_delta0": true,  
    "delta0_fallback": -0.0001,  
    "g": 0.891,  
    "L_eff": 2.22,  
    "K_us": 0.0015,  
    "tau": 0.069  
  },  
  "HYUNDAI_IONIQ_5": {  
    "use_per_segment_delta0": true,  
    "delta0_fallback": 0.0,  
    "g": 0.938,  
    "L_eff": 2.887,  
    "K_us": 0.00289,  
    "tau": 0.062  
  }  
}
```

```
}  
}
```

## module-3.v2\_agent-10

```
{  
  "FORD_F_150_LIGHTNING_MK1": {  
    "use_per_segment_delta0": false,  
    "delta0": 0.00133,  
    "g": 0.863,  
    "L_eff": 3.26,  
    "K_us": 0.0035,  
    "tau": 0.06  
  },  
  "FORD_MUSTANG_MACH_E_MK1": {  
    "use_per_segment_delta0": true,  
    "delta0_fallback": -0.0001,  
    "g": 0.891,  
    "L_eff": 2.22,  
    "K_us": 0.0015,  
    "tau": 0.069  
  },  
  "HYUNDAI_IONIQ_5": {  
    "use_per_segment_delta0": true,  
    "delta0_fallback": 0.0,  
    "g": 0.938,  
    "L_eff": 2.887,  
    "K_us": 0.00289,  
    "tau": 0.062  
  }  
}
```

## module-3.v3\_agent-02

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "a": 0.9839278434685126,
    "b": -0.0004596324959947,
    "c": 0.0003973122452781526,
    "d": -0.007997476152353435
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "a": 0.9727773793773095,
    "b": 0.0016981202953789607,
    "c": 0.0001999394762639504,
    "d": -0.02207103010528879
  },
  "HYUNDAI_IONIQ_5": {
    "a": 0.9913665028013989,
    "b": 0.0006388079681906442,
    "c": 0.00014836133647526484,
    "d": 0.00450672562112523
  }
}
```

## module-3.v3\_agent-06

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "w": [
      0.039210004724697683,
      0.0027898060882411775,
      -8.213353932869296e-05,
      -0.002101385572330489,
      0.015375393428030156,
      -0.019248388299671407,
      0.0017437669476709928
    ],
    "features": [
      "yr_v1",
      "abs_yr_v1",
      "v",
      "v*yr_v1",
      "ddelta_dt",
      "delta",
      "1"
    ],
    "lambda": 30.0
  }
}
```

```
},
"FORD_MUSTANG_MACH_E_MK1": {
  "w": [
    0.04786803033956984,
    0.0010252169296772547,
    -0.0001550342759403376,
    0.0007801849572229949,
    0.03973697702371263,
    -0.04132919662285873,
    0.0029539513418850413
  ],
  "features": [
    "yr_v1",
    "abs_yr_v1",
    "v",
    "v*yr_v1",
    "ddelta_dt",
    "delta",
    "1"
  ],
  "lambda": 30.0
},
"HYUNDAI_IONIQ_5": {
  "w": [
    0.11163643144945354,
    -0.01144073312700024,
    -8.938098452332296e-05,
    -0.0043289766474131726,
    0.011781066593348639,
    -0.1397503946290344,
    0.0011078590121685729
  ],
  "features": [
    "yr_v1",
    "abs_yr_v1",
    "v",
    "v*yr_v1",
    "ddelta_dt",
    "delta",
    "1"
  ],
  "lambda": 30.0
}
```

## module-3.v3\_agent-08

```
{
  "FORD_F_150_LIGHTNING_MK1": {
    "features": [
      "bias",
      "delta_road",
      "v",
      "v_delta",
      "ddelta",
      "v_ddelta",
      "v0_yaw",
      "abs_v0_yaw",
      "a_long",
      "v_sq_delta"
    ],
    "beta": [
      -0.0013823839722696969,
      0.04137416440068437,
      7.448672929471383e-05,
      -0.012310248841363941,
      -0.006256706876683462,
      -0.0013792677587728672,
      0.0045077768608475025,
      -0.004229129267950782,
      0.00046053718010623685,
      0.00047676588153397785
    ],
    "n_samples": 430990
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "features": [
      "bias",
      "delta_road",
      "v",
      "v_delta",
      "ddelta",
```

```

    "v_ddelta",
    "v0_yaw",
    "abs_v0_yaw",
    "a_long",
    "v_sq_delta"
  ],
  "beta": [
    -0.0002765413679144323,
    0.04444765601029079,
    7.361683614844277e-05,
    -0.010525182430347446,
    0.0006366996373336289,
    -0.003027149663796255,
    0.004991171014731363,
    0.01051162443915557,
    -0.0003707902497743102,
    -0.0004177085900277482
  ],
  "n_samples": 615175
},
"HYUNDAI_IONIQ_5": {
  "features": [
    "bias",
    "delta_road"
  ]
}

```

## module-3.v3\_agent-10

```

{
  "FORD_F_150_LIGHTNING_MK1": {
    "intercept": -0.0015292016694573706,
    "weights": {
      "v_mps": 7.523513452594086e-05,
      "delta_road_rad": 0.015247619064453013,
      "d_delta_dt": -0.008088882179724404,
      "a_long_mps2": 0.0004266683248191665,
      "yr_v1": -0.010938174180205732,
      "abs_delta": -0.0046619127252169905,
      "yr_v1_sq_signed": 0.0031131630350753935
    }
  }
}

```

```

    }
  },
  "FORD_MUSTANG_MACH_E_MK1": {
    "intercept": -0.00025671546619143265,
    "weights": {
      "v_mps": 7.597710383706725e-05,
      "delta_road_rad": 0.02230237156926512,
      "d_delta_dt": -0.004979903507745553,
      "a_long_mps2": -0.0003707521334163227,
      "yr_v1": -0.025822216062859112,
      "abs_delta": 0.022243029033033466,
      "yr_v1_sq_signed": 0.012709323274160847
    }
  },
  "HYUNDAI_IONIQ_5": {
    "intercept": -7.400008140931345e-06,
    "weights": {
      "v_mps": 3.4765997046644015e-05,
      "delta_road_rad": 0.007171887290213692,
      "d_delta_dt": 9.749220428357943e-05,
      "a_long_mps2": 0.0006747858420646872,
      "yr_v1": -0.010175331098695933,
      "abs_delta": 0.009304240486557539,
      "yr_v1_sq_signed": 0.00642116668196153
    }
  }
}

```

## - 6b. Operating-contract audit



The grader strips `sim_df` to an allowlist before calling each agent's `predict()`, so an agent cannot read truth at inference time. This section reports source-level intent — which agents *would have* read truth columns if the grader hadn't intervened. A clean audit is its own quality signal.

**Clean cohort:** no agent referenced truth-derived column names inside their `predict()` body.

## 7. Reconstruction quality (substrate signal)

How many agents shipped the right artefacts to be canonically gradable. Failures here are a substrate or contract problem, not a model problem.

| format check                       | pass | fail |
|------------------------------------|------|------|
| agent_folder_exists                | 70   | 0    |
| has_manifest_json                  | 70   | 0    |
| manifest_parsable                  | 70   | 0    |
| manifest_declares_predict_callable | 70   | 0    |
| manifest_declares_platform_support | 70   | 0    |

| format check    | pass | fail |
|-----------------|------|------|
| has_predict_py  | 70   | 0    |
| has_coeffs_json | 61   | 9    |
| has_report      | 33   | 37   |

#### Failure reasons across the cohort:

- import\_failed — 1 agent(s)

## • 8. Worst-of-cohort

### Lowest yaw $\Delta\%$

- m1-agent-08  
(+40.4%)
- m2-agent-04  
(+44.6%)
- module-2.v2\_agent-03  
(+45.3%)

### Lowest CTE $\Delta\%$

- m1-agent-10  
(+49.5%)
- m2-agent-07  
(+49.5%)
- m2-agent-04  
(+50.0%)

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lateral-attribution